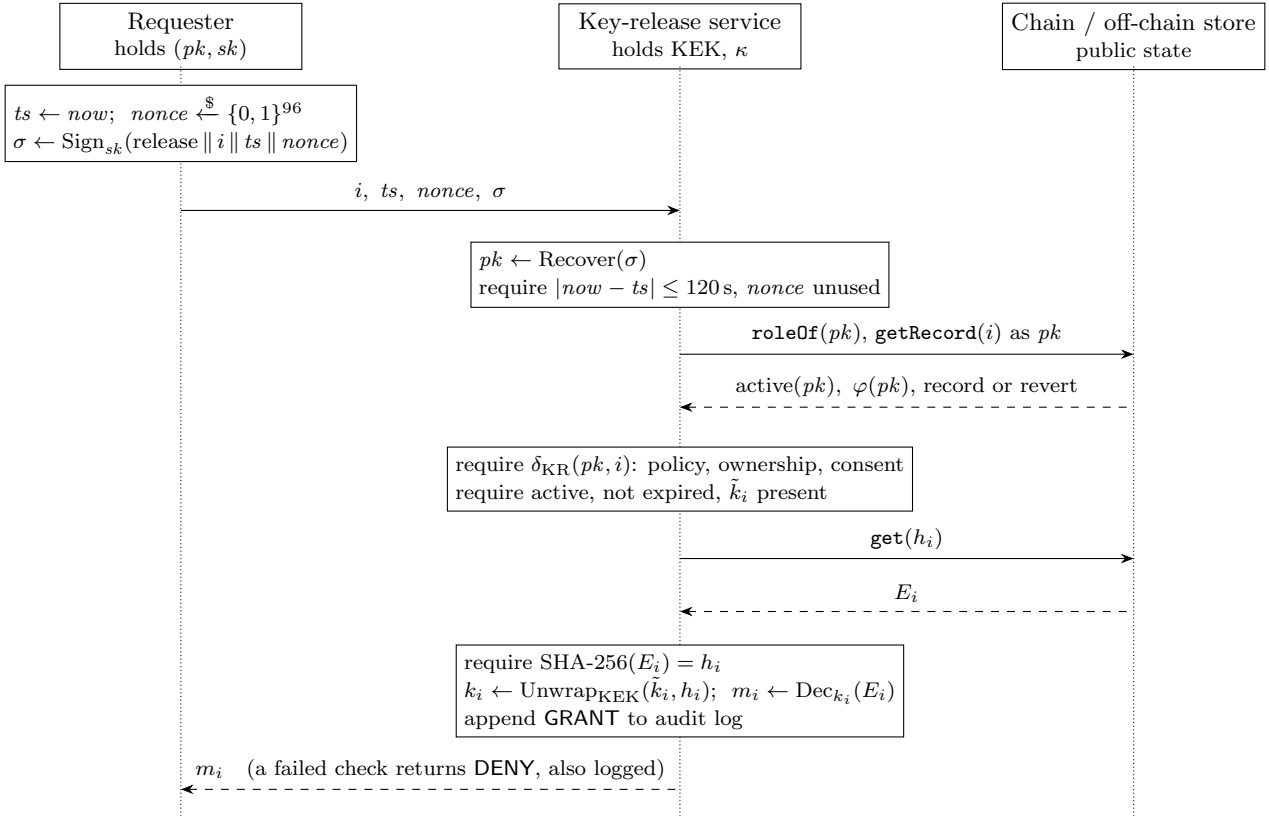

Requester
holds (pk, sk)
Key-release service
holds KEK, κ
Chain / off-chain store
public state
$ts \leftarrow now$; $nonce \xleftarrow{\$} \{0,1\}^{96}$
$\sigma \leftarrow \mathrm{Sign}_{sk}(\mathrm{release} \,\|\, i \,\|\, ts \,\|\, nonce)$
i, ts, $nonce$, σ
$pk \leftarrow \mathrm{Recover}(\sigma)$
require $|now - ts| \leq 120\,\mathrm{s}$, $nonce$ unused
roleOf(pk), getRecord(i) as pk
active(pk), $\varphi(pk)$, record or revert
require $\delta_{\mathrm{KR}}(pk, i)$: policy, ownership, consent
require active, not expired, $\tilde{k}_i$ present
get(h_i)
E_i
require SHA-256$(E_i) = h_i$
$k_i \leftarrow \mathrm{Unwrap}_{\mathrm{KEK}}(\tilde{k}_i, h_i)$; $m_i \leftarrow \mathrm{Dec}_{k_i}(E_i)$
append GRANT to audit log
m_i (a failed check returns DENY, also logged)